Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item

Instructions

1. Which of the following assertions accurately characterize adversarial attacks? (Select all that apply.) 1 / 1 point

- [ ] They function as a type of data encryption.
- [ ] They invariably necessitate complex algorithms for execution.
- [x] They serve as a mechanism to evaluate and enhance model robustness.

Nice work

Correct! Adversarial attacks are often employed to identify and address vulnerabilities, which can lead to stronger and more resilient machine learning models.

- [ ] They are exclusively employed for nefarious purposes.
- [x] They leverage vulnerabilities inherent in machine learning models.

Nice work

Correct! Adversarial attacks exploit weaknesses within machine learning models to influence their predictions or decisions.

2. How does Generative Adversarial Network (GAN) logic facilitate the circumvention of network anomaly detectors? 1 / 1 point

- ( ) By encrypting malicious traffic to render it unidentifiable
- ( ) By launching direct attacks against the anomaly detection algorithms
- (•) By synthesizing data that emulates typical network traffic patterns
- ( ) By modifying the protocols governing data transmission

Nice work

Correct! GANs generate data that mirrors legitimate traffic, making it challenging for anomaly detection systems to identify malicious activity.

3. Imagine a scenario where a financial institution uses machine learning models to detect fraudulent transactions in real-time. An attacker attempts to exploit this system by slightly modifying the transaction attributes to bypass the fraud detection model. Which type of adversarial attack is the attacker most likely employing, and how does it function? 1 / 1 point

- (•) Fast Gradient Sign Method (FGSM), by slightly perturbing the input data to mislead the model
- ( ) Autoencoder-based attack, by compressing the transaction features.
- ( ) Latent space manipulation, by altering deep features
- ( ) Reinforcement learning attack, by training the model to ignore fraudulent behavior

Nice work

Correct! FGSM introduces small, calculated changes to transaction attributes, aimed at tricking the fraud detection system into making incorrect predictions without raising suspicion.

4. What defines a fundamental characteristic of a Projected Gradient Descent (PGD) attack? 1 / 1 point

- ( ) It seeks to minimize perturbations to remain undetectable
- ( ) It alters input data through a single-step process
- ( ) It enhances a model's defenses against adversarial attacks
- (•) It incrementally perturbs input data within a predefined boundary

Nice work

Correct! PGD operates by iteratively altering the input data within a designated range to deceive the neural network.

5. What is the primary goal of a Fast Gradient Sign Method (FGSM) attack? 1 / 1 point

- ( ) To enhance the robustness of neural networks
- (•) To introduce minor perturbations in the input data to deceive the neural network
- ( ) To train a neural network to identify adversarial attacks
- ( ) To encrypt data as a safeguard against unauthorized access

Nice work

Correct! The FGSM method perturbs input data slightly in order to mislead the neural network's prediction.